[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION
GRINDING MACHINE, TOOL AND CUTTER, COMPUTER NUMERICAL CONTROL (CNC)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers computer numerical control (CNC) tool and cutter grinding machines. The machines are intended for grinding and resharpening conventional cutters, including millers, reamers, counterbores, hobs, and other metal-cutting tools. The tools may have cylindrical or tapered sides; straight or helical flutes; ball, radius, or square noses; and right- or left-hand spirals.

2. CLASSIFICATION. The grinding machines shall be of the following classes and sizes. The class and size to be furnished shall be as specified (see 7.2(b)).

Class 1 - two CNC axes
Class 2 - three CNC axes
Class 3 - four CNC axes
Class 4 - five CNC axes

Size - Sizes are shown in table I

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3415

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TABLE I. Sizes of grinding machines.

Size	Class	Workpiece dimensions, inches		Configuration of CNC axes						
				Linear			Workhead		Wheelhead	
		Swing	Length	X-axis	Y-axis	Z-axis	A-axis	C-axis	B-axis	C-axis
1	1	8	6	X			X			
2		10	10	X			X			
3		10	12	X			X			
4		10	36	X			X			
5		11	14	X			X			
6		12	14	X			X			
7	2	2.5	18	X	X		X			
8		6	4	X	X		X			
9		8	18	X	X		X			
10		11	10	X	X		X			
11		11	14	X	X		X			
12		12	12	X	X		X			
13		16	28	X	X		X			
14	3	1.5	8	X	X	X	X			
15		11	16	X	X	X	X			
16		24	36	X	X	X	X			
17	4	4	8	X	X	X	X	X		
18		4	10	X	X	X	X	X		
19		4	10	X	X	X	X			X
20		4	24	X	X	X	X		X	
21		6	24	X	X	X	X		X	
22		8	10	X	X	X	X	X		
23		8	12	X	X	X	X	X		
24		9.5	14	X	X	X	X	X		
25		12	12	X	X	X	X	X		
26		14	26	X	X	X	X			X
27		18	18	X	X	X	X	X		
28		18	18	X	X	X	X		X	

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3 SALIENT CHARACTERISTICS

3.1 General requirements. The grinding machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.1.1 Identification of CNC axes. The longitudinal (X-axis) feed and infeed (Y-axis) shall be provided by either workhead table or wheelhead motion. The vertical (Z-axis) feed shall be provided by wheelhead motion. When workhead rotational axes are provided, the workhead shall rotate about a horizontal A-axis and the workhead or compound rotary table shall rotate about a vertical C-axis. When a wheelhead rotational axis is provided, the wheelhead shall rotate about a vertical B-axis or about a horizontal C-axis.

3.2 Components. The grinding machine shall include, as a minimum, the following components:

3.2.1 Base and bed. The base shall support other elements of the machine and incorporate materials, shape, and mass to minimize adverse vibration and deflection. The bed shall be an integral part of the base and provide the surface required for machineways and mounting pads. The base shall be provided with a means for leveling and anchoring the machine to the floor.

3.2.2 Worktable. The worktable shall provide rigid support for mounting the workhead and, when provided, the tailstock.

3.2.3 Machineways. Machineways shall be of the design considered by the manufacturer as standard for the machine. All machineways shall have protective covers and/or wipers to keep swarf and other debris from bearing surfaces.

3.2.4 Workhead. The workhead assembly shall consist of a support frame, workhead spindle, and spindle drive mechanism.

3.2.4.1 Workhead spindle. Unless otherwise specified (see 7.2(c)), the workhead spindle shall have a taper for use with centers, chucks, and taper shank workpieces that is considered by the manufacturer as standard. The workhead spindle shall be mounted on antifriction or low-friction bearings that are automatically lubricated or sealed for life and designed to minimize axial and radial runout. Unless otherwise specified (see 7.2(d)), the CNC workhead spindle shall have synchronous rotation (A-axis) in both directions and incremental rotary indexing of 0.001 degree.

3.2.5 Tailstock. Unless otherwise specified (see 7.2(e)), the class 1, 2, or 3 grinding machine shall be provided with a tailstock considered by the manufacturer as standard.

3.2.6 Wheelhead assembly. The wheelhead assembly shall consist of a support frame, wheel spindle(s), wheel spindle drive mechanism(s), grinding wheel(s), and protective cowling. If required, the wheelhead assembly have multiple grinding wheel spindles mounted individually or on a turret as specified (see 7.2(f)).

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3.2.6.1 Grinding wheel spindle(s). The grinding wheel spindle(s) shall be supported and held in axial and radial alignment by precision antifriction bearings. Continuous or permanent lubrication of the spindle bearings shall be provided. The spindle(s) shall be balanced to provide vibration-free operation at all spindle speeds. When specified (see 7.2(g)), the spindle shall accept premounted and balanced grinding wheels. Unless otherwise specified (see 7.2(h)), the spindle drive system, motor horsepower, and spindle speed shall be that considered by the manufacturer as standard for the grinding machine.

3.2.6.2 Grinding wheel(s). Grinding wheels shall be supplied with the grinding machine as specified (see 7.2(i)). Unless otherwise specified (see 7.2(j)), grinding wheels shall conform to the shapes and sizes of American National Standards Institute (ANSI) B74.2, be marked in accordance with ANSI B74.13, and meet safety requirements of ANSI B7.1.

3.2.7 Stops. Stops shall be provided on all linear axes of motion to prevent damage to the grinding machine from overtravel.

3.2.8 CNC system. The CNC system shall be based on either a personal computer (PC) open-architecture or other software programmable microprocessor design. The CNC system shall be industrial grade in accordance with accepted industry practices and shall meet Government requirements as specified (see 7.2(k)).

3.2.8.1 Hardware features. Unless otherwise specified (see 7.2(l)), the CNC system shall include, as a minimum, the following hardware:

a. One or more 32-bit microprocessors with math coprocessor.
b. 250-MB hard disk drive.
c. 3.5-inch high density floppy disk drive.
d. DNC link capability.
e. 9-inch monochrome monitor.
f. Manual data input (MDI) device: PC keyboard or alphanumeric keyboard with dedicated keys for specific machine functions.
g. Dual Electronic Industries Association (EIA) 232 communication ports.

3.2.8.2 Control panel. The operator controls, display panel, alarms, and indicators shall be located at the operator's workstation and shall be convenient to and in full view of the operator during setup and operation. Unless otherwise specified (see 7.2(m)), the CNC system shall include, as a minimum, a control panel with the following manual controls and switches:

a. System keyed lock.
b. Emergency stop button.
c. Power on/off switches.
d. CNC system start/pause/stop switches.
e. Display of operator alerts and alarms.
f. Display of active program progress.
g. Manual override of active program.

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h. Variable infeed.
i. Feed hold.
j. Feed rate override rotary switches.
k. Variable jog and jog direction.

3.2.8.3 Operating modes. Unless otherwise specified (see 7.2(n)), the CNC system shall include, as a minimum, the following operating modes:

a. Manual operation mode.
b. Manual data input mode (MDI).
c. Setup mode.
d. Learn mode.
e. Transmit mode.
f. Automatic mode.
g. Stored program editing.
h. Background editing/split screen program preview.
i. Machine diagnostic/maintenance/monitoring.

3.2.8.4 Axis and tool management. Unless otherwise specified (see 7.2(o)), the CNC system shall include, as a minimum, the following axis and tool management features:

a. Grinding wheel diameter compensation.
b. Grinding wheel breakage detection.
c. Grinding wheel offsets.
d. Scale feedback.
e. Backlash compensation.
f. Work zone/motion collision/limit detection.
g. Repeatable error correction/compensation.

3.2.8.5 Software features. Unless otherwise specified (see 7.2(p)), the CNC system shall include, as a minimum, the following software programming and control features:

a. Simultaneous control of all CNC axes.
b. Linear, circular, parabolic, and helical interpolation.
c. Inch/metric switchable programming.
d. Decimal point programming.
e. Workpiece graphic display rotation.
f. Tool path graphic display.
g. Standard EIA 494 codes compatability.
h. Dimensional absolute/incremental programming.
i. Cartesian/polar coordinate programming.
j. Programmable zero reference.

3.2.9 Lubrication system. Unless otherwise specified (see 7.2(q)), the grinding machine shall be equipped with an automatic lubricating system to direct oil flow from the reservoir to all machine

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components requiring continuous lubrication. A means shall be provided for determining oil level in the reservoir. The automatic lubricating system shall include a cleanable or replaceable filter. Upon loss of pressure in the automatic lubrication system, the grinding machine shall automatically stop. Machine parts requiring periodic manual lubrication shall be readily accessible through doors or removable panels.

3.2.10 Coolant system. Unless otherwise specified (see 7.2(r)), the grinding wheel shall be flood cooled. As a minimum, the coolant system shall consist of a reservoir, pump, strainer, directional nozzles, and controls. The system shall be equipped with a method to control the flow rate, deliver clean coolant to the work zone, and collect spent coolant from the work zone. Unless otherwise specified (see 7.2(s)), a coolant filtering system that is considered by the manufacturer as standard for the machine shall be provided.

3.2.11 Hydraulic system. If provided, the hydraulic system shall conform to the requirements of National Fluid Power Association (NFPA) T2.24.1. The hydraulic system shall be complete and include a reservoir, pump, valves, piping, cylinders, fluid level indicator, and controls. A means shall be provided to prevent damage to components from overpressure. A filter system shall be provided to ensure delivery of clean fluid.

3.2.12 Electrical system. Unless otherwise specified (see 7.2(t)), the electrical system shall conform to the National Fire Protection Association (NFPA) 79. The grinding machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(u)). Unless otherwise specified (see 7.2(v)), the CNC system power shall be provided by a separate 120-volt, single-phase, 60-hertz circuit from the grinding machine main power supply.

3.2.13 Electric motors. The grinding machine drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG 1. Unless otherwise specified (see 7.2(w)), the motors shall have open dripproof enclosures.

3.2.14 Optional accessories. Optional accessories shall be provided as specified (see 7.2(x)).

3.3 Performance. Unless otherwise specified (see 7.2(y)), the grinding machine shall meet the alignment and performance requirements shown in tables II and III, respectively, when tested in accordance with sections 5.3 through 5.3.5.

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TABLE II. Alignment requirements.

Parameter	Tolerance
Worktable surface flatness	±0.0002 inch/foot
Worktable surface, square to infeed travel of wheelhead	±0.0005 inch/foot
Longitudinal travel, deviation between worktable surface and spindle	±0.0005 inch/foot
Infeed travel, deviation between worktable surface and spindle	±0.0005 inch/foot
Spindle axial endplay and radial runout	0.0002 inch TIR[1]
Rotary table surface, parallel with linear travel over 90 degrees rotation	±0.0005 inch/foot

[1] TIR is total indicator reading.

TABLE III. Performance requirements.

Parameter	Tolerance
Linear axis positioning accuracy	±0.0004 inch/foot
Linear axis positioning repeatability	±0.0002 inch/foot
Maximum leadscrew reversal error (software compensated)	0.0001 inch
CNC angular axis positioning accuracy	±0.001 degree
CNC angular axis positioning repeatability	±0.0005 degree
Workpiece dimensional accuracy	±0.0005 inch
Workpiece surface roughness, Rt[1]	16 microinches

[1] Rt is roughness total as defined in American Society of Mechanical Engineers (ASME) B46.1.

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(z)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.9. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI B11.9 and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(aa)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(bb)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(cc)), the nameplate shall contain the following information:

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a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(dd)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The grinding machine shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The grinding machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(ee)), the grinding machine shall be tested in accordance with sections 5.3.1 through 5.3.5.

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5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(ff)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(gg)). Unless otherwise specified (see 7.2(hh)), the machine shall be tested in the environmental conditions of the manufacturer's facility.

5.3.2 Operational test. The grinding machine shall be operated in accordance with the manufacturer's standard operating procedures. The grinding machine shall be operated at no load for not less than 4 hours. Proper operation of all controls and features shall be verified during the trial period, after which the initial alignment test shall be conducted.

5.3.3 Alignment tests. The grinding machine alignment tests shall be conducted before and after the performance tests and shall meet the requirements shown in table II. Changes in alignment shall be cause for rejection.

5.3.4 Performance test. The performance test shall be conducted in accordance with the manufacturer's standard operating procedures. Unless otherwise specified (see 7.2(ii)), the performance test shall consist of automatically grinding a cutter selected by the Government from ANSI B94.19. The workpiece shall be formed from a plain blank and finished in accordance with ANSI B94.19 for the cutter selected. Five identical workpieces shall be ground using the same CNC machining program. All workpieces shall meet the requirements of ANSI B94.19 and the performance requirements shown in table III. Test workpieces shall match each other at any geometric point with variations no greater than 0.001 inch. Selected surfaces on the finished workpieces shall have a total roughness (Rt) no greater than 16 microinches. The radial runout of the finished workpieces shall not be greater than 0.0006 inch total indicator reading (TIR).

5.3.5 Optional tests. Optional tests shall be conducted as specified (see 7.2(jj)).

5.4 Acceptance. Unless otherwise specified (see 7.2(kk)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the tests shall be compared with the alignment and performance requirements shown in tables II and III, respectively. Failure of the grinding machine to meet the requirements shown in tables II and III shall be cause for rejection.

6. PACKAGING. Unless otherwise specified (see 7.2(ll)), preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Sources of referenced documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B7.1	Safety Requirements for the Use, Care and Protection of Abrasive Wheels
ANSI B11.9	Safety Requirements for the Construction, Care, and Use of Grinding Machines
ANSI B74.13	Markings for Identifying Grinding Wheels and Other Bonded Abrasives
ANSI B74.2	Specifications for Shapes and Sizes of Grinding Wheels and for Shapes, Sizes, and Identification of Mounted Wheels
ANSI B94.19	Milling Cutters and End Mills

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

American Society of Mechanical Engineers (ASME)

ASME B46.1	Classification and Designation of Surface Qualities

Applications for copies should be sent to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

Electronic Industries Association (EIA)

EIA 232	Interface Between Data Terminal Equipment and Data Circuit-Terminating Equipment Employing Serial Binary Data Interchange
EIA 494	32 Bit Binary CL (BCL) and 7 Bit ASCII CL (ACL) Exchange Input Format for Numerically Controlled Machines

Applications for copies should be sent to the Electronic Industries Association, 2500 Wilson Boulevard, Arlington, VA 22201-3834.

National Electrical Manufacturers Association (NEMA)

NEMA MG 1	Motors and Generators

Applications for copies should be sent to the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

National Fire Protection Association (NFPA)

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NFPA 79 Electrical Standards for Industrial Machinery

Applications for copies should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

National Fluid Power Association (NFPA)

NFPA T2.24.1 Hydraulic Fluid Power - Systems Standard for Stationary Industrial Machinery

Applications for copies should be sent to the National Fluid Power Association, 3333 North Mayfair Road, Milwaukee, WI 53222-3219.

Occupational Safety and Health Association (OSHA)

OSHA 1910 Occupational Safety and Health Standards for General Industry (29 CFR 1910)

Applications for copies should be sent to the U. S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Class and size required (see 2).
c. Workhead spindle taper, if different (see 3.2.4.1).
d. Workhead spindle angular rotation, if different (see 3.2.4.1).
e. Tailstock, if different (see 3.2.5).
f. Multiple grinding wheel spindles, as specified (see 3.2.6).
g. Grinding wheel spindle for premounted wheels, if required (see 3.2.6.1).
h. Grinding wheel spindle drive system, horsepower, and speed, if different (see 3.2.6.1).
i. Grinding wheels, as specified (see 3.2.6.2).
j. Grinding wheel standards, if different (see 3.2.6.2).
k. CNC system, as specified (see 3.2.8).
l. CNC system hardware, if different (see 3.2.8.1).
m. CNC system control panel, if different (see 3.2.8.2).
n. CNC system operating modes, if different (see 3.2.8.3).
o. CNC system axis and tool management, if different (see 3.2.8.4).
p. CNC system software, if different (see 3.2.8.5).
q. Lubrication system, if different (see 3.2.9).
r. Coolant system, if different (see 3.2.10).
s. Coolant filtering system, if different (see 3.2.10).
t. Electrical system industrial standard, if different (see 3.2.12).
u. Electrical system voltage, as specified (see 3.2.12).

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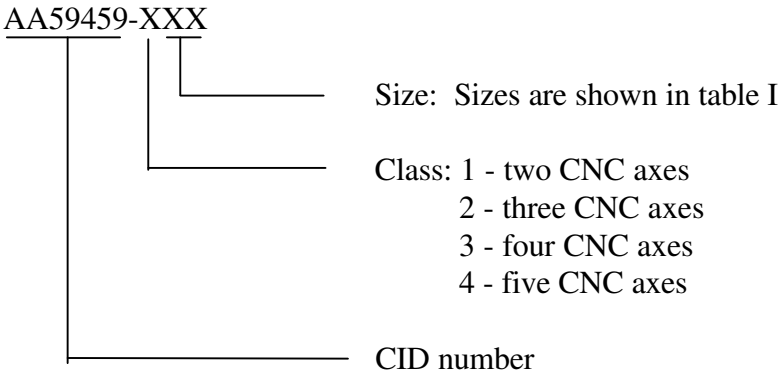
v. CNC system power, if different (see 3.2.12).
w. Motor enclosures, if different (see 3.2.13).
x. Optional accessories, as specified (see 3.2.14).
y. Performance, if different (see 3.3).
z. Machine dimensions and weight, as specified (see 3.4).
aa. System of units required (see 3.6).
bb. Dual system of units, if required (see 3.6).
cc. Nameplate, if different (see 3.7).
dd. Lubrication plate or chart, if required (see 3.8).
ee. Testing, if different (see 5.3).
ff. Test responsibility, if different (see 5.3.1).
gg. Certification of calibration, as specified (5.3.1).
hh. Test environmental conditions, if different (see 5.3.1).
ii. Performance test, if different (see 5.3.4).
jj. Optional tests, as specified (see 5.3.5).
kk. Acceptance, if different (see 5.4).
ll. Packaging requirements, as specified (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



MILITARY INTERESTS:

Custodians
Air Force - 99
Army - AL
Navy - SH

Reviewers
Air Force - 84
Army - AR, CR4, SM
Navy - MC

PREPARING ACTIVITY:
DLA - GS

(Project 3415-0255)

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